

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 12 — Environmental Chemistry: Water

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Anhydrous copper(II) sulphate turns ____ when it detects water. ✓ **B — blue** 1 mark

Anhydrous copper(II) sulphate is a **white crystalline solid**. On absorbing water it becomes **hydrated copper(II) sulphate**, which is blue. This colour change from white to blue confirms the presence of water. Options A, C, and D describe colours that do not occur in this reaction.

Q2. The melting point of pure water is: ✓ **A — 0°C** 1 mark

Pure water **freezes at 0°C** and boils at 100°C at sea level. These are standard reference values. **Impurities lower the melting point** below 0°C, which is how the purity test works. 4°C is the temperature of maximum density of water, not its melting point.

Q3. The pH of distilled water is: ✓ **C — 7** 1 mark

Distilled water is free of all contaminants and has a **neutral pH of 7**. This is stated directly in Section 12.2.1. Tap water may have a slightly different pH because of dissolved minerals and treatment chemicals.

Q4. Substance added during the coagulation step: ✓ **B — slaked lime and alum** 1 mark

In **coagulation**, slaked lime Ca(OH)_2 and alum $\text{Al}_2(\text{SO}_4)_3$ react to form a gelatinous mass of **aluminium hydroxide Al(OH)_3** , which carries down dirt particles and bacteria. Chlorine is added in the later chlorination step, and sand/gravel are used in filtration.

Q5. Chlorination kills: ✓ **B — bacteria** 1 mark

Chlorine reacts with water to form **hypochlorous acid HClO** , which is bactericidal. Heavy metals, phosphates, and nitrates are not removed by chlorination; they require physical or other chemical processes. Note: chlorination is not effective against *viruses*.

Q6. Waterborne disease caused by *entamoeba*: ✓ **C — Dysentery** 1 mark

Dysentery is caused by the parasite **entamoeba**. It spreads through faecal contamination of water or food. Typhoid is caused by *salmonella typhi* bacteria, and hepatitis is caused by viruses. Jaundice is a symptom of liver disease, not a disease caused by *entamoeba*.

Q7. Which is NOT a waterborne disease? ✓ **D — Anaemia** 1 mark

Anaemia is a blood condition caused by iron deficiency, not by contaminated water. The textbook lists **dysentery, hepatitis, jaundice, and typhoid** as waterborne diseases in Section 12.4. This question is drawn directly from Review Question 1(i).

Q8. High concentrations of phosphates cause: ✓ **B — Eutrophication** 1 mark

Eutrophication is excessive algae growth in water bodies driven by phosphate and nitrate runoff from fertilizers and detergents. The algae deplete oxygen, killing aquatic life. Chlorination and sedimentation are water treatment steps, not consequences of phosphate pollution.

Q9. NPK fertilizers provide: ✓ **B — N, P, K** 1 mark

NPK stands for **nitrogen (N), phosphorus (P), and potassium (K)**. These three elements are the main nutrients that fertilizers provide to enhance plant growth. Na (sodium) and Pb (lead) are not plant nutrients in this context; Ca (calcium) is a secondary nutrient, not one of the NPK trio.

Q10. Human activity that causes water contamination: ✓ **D — All of these** 1 mark

Section 12.3 and Review Question 1(ii) confirm that **livestock waste, pesticides, and septic tanks** all contaminate water bodies. The correct answer is "all of these" because the textbook lists all three explicitly among sources of water pollution.

Q11. Distilled water is preferred in labs because: ✓ **B — it is free of contaminants** 1 mark

Distilled water has undergone **distillation** to remove all dissolved solids and contaminants. Its **electrical conductivity is very low** (not high), and its pH is 7 (neutral). For laboratory reagents and calibration of analytical instruments, absence of contaminants is essential.

Q12. Water scarcity in Pakistan is mainly caused by: ✓ **B — population growth, climate change, and inadequate supply** 1 mark

Section 12.6 states three causes: (1) **rapid population growth** increasing demand, (2) **climate change** causing irregular rainfall and declining aquifers, and (3) **inadequate water supply** from traditional flood irrigation leading to large evaporation losses. The other options are not causes of water scarcity.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(ii) Activity 12.1: Detection of water with anhydrous copper(II) sulphate 3 marks

Procedure: A small amount of anhydrous copper(II) sulphate powder (white) is placed in a dry test tube. A few drops of the water sample are added with a dropper. 1 mark

Colour change: The white powder turns **light blue**, confirming the presence of water. 1 mark

Equation: CuSO_4 (white) + $5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue). The product is hydrated copper(II) sulphate. 1 mark

OR

Melting point purity test: Ice cubes are made from the sample water and placed in a beaker. A thermometer is immersed and the ice is heated gradually. The temperature at which ice begins to melt is recorded. 1 mark

Standard value: Pure water melts at exactly **0°C**. 1 mark

Interpretation: If the observed melting point is **below 0°C**, it indicates the presence of dissolved impurities (solutes depress the melting point), so the water is impure. 1 mark

2(ii) Properties of pure water / Impurities in natural water 3 marks

Physical properties of pure water:

- Transparent, colourless, odourless, and tasteless. 1 mark
- Boiling point: **100°C** at sea level. 1 mark
- Freezing (melting) point: **0°C** at sea level. 1 mark

OR

Natural water is not pure because **water is an excellent solvent**. As rainwater falls through the atmosphere it dissolves: **oxygen, nitrogen, carbon dioxide, and dust particles**. During thunderstorms it also dissolves nitric acid. 1 mark each for any four substances, up to 3 marks

Groundwater additionally dissolves minerals from rocks and soils and substances from decaying organisms.

2(iii) Harmful substances in natural water 3 marks

Any three of the following, one mark each:

- **Plastics:** harm aquatic life and disrupt ecosystems. 1 mark
- **Wastewater/sewage:** carries bacteria, viruses, and pathogens that cause disease in humans and aquatic life. 1 mark
- **Metal compounds (toxic metals):** excessive concentration is poisonous to aquatic organisms and humans. 1 mark
- **Nitrates from fertilizers:** high concentrations in drinking water cause health risks, especially for young children. 1 mark
- **Phosphates from fertilizers/detergents:** cause eutrophication, harmful algal blooms, and oxygen depletion. 1 mark

OR

Beneficial substances in natural water:

- **Dissolved oxygen:** aquatic life depends on it for respiration; supports survival of fish and other organisms. 1 mark
- **Essential minerals (metal compounds):** iron is needed for haemoglobin production; zinc, copper, and manganese activate enzyme activity for biological processes. 2 marks

2(iv) Steps of raw water treatment 3 marks

The four steps in order, one mark for any three correctly stated:

1. **Sedimentation:** water stands in a reservoir; suspended matter sinks to the bottom. ½ mark
 2. **Coagulation:** slaked lime and alum are added; a gelatinous $\text{Al}(\text{OH})_3$ mass forms and carries down dirt and bacteria. ½ mark
 3. **Filtration:** water passes through sand and gravel (sometimes charcoal) to remove coloured and odorous compounds. ½ mark
 4. **Chlorination:** chlorine is added to kill remaining bacteria by forming HClO . ½ mark
- (Award 1 mark per step correctly described, up to 3 marks.)

OR

Coagulation equation:



Product: $\text{Al}(\text{OH})_3$ is a **gelatinous (jelly-like) mass** of aluminium hydroxide. It traps and carries down fine dirt particles and bacteria as it settles, making the water clearer. 1 mark

2(v) Chlorination equation and mechanism 3 marks



Acid produced: Hypochlorous acid (HClO). 1 mark

Mechanism: HClO is a powerful disinfectant; it kills any remaining bacteria in the water, making it safe to drink. This is the final step in the purification process. 1 mark

OR

Limitation of chlorination: Chlorination kills **bacteria** but is **not effective against viruses** such as the hepatitis virus. 1 mark

This is because viruses have a different structure from bacteria and HClO cannot destroy them at the concentrations used in water treatment. 1 mark

Alternative treatment: Viral contamination requires additional methods such as **UV irradiation** or **boiling** the water, which physically destroy virus particles. 1 mark

2(vi) Distilled water vs. tap water 3 marks

Aspect	Distilled Water	Tap Water
Source	Produced by distillation (boiling and condensing)	Municipal system from streams, rivers, lakes, groundwater
Purity	Free of all contaminants; no dissolved solids	Treated but still contains dissolved minerals and trace chemicals
Lab use	Preparing chemical reagents and solutions; calibrating instruments (e.g. pH meter)	General cleaning of glassware; educational demonstrations

1 mark per row, up to 3 marks

OR

Long-term drinking of distilled water is not advisable because it **lacks essential minerals** found in natural water sources. 1 mark

The textbook "Do You Know?" box states: long-term drinking of distilled water can cause **mineral deficiency** because it lacks calcium, magnesium, and other minerals essential for biological processes.

1 mark

For drinking purposes, it is better to use **treated tap water, bottled water, or natural spring water** which retain these essential minerals. 1 mark

2(vii) Water pollution: definition and causes 3 marks

Definition: Water pollution is the **contamination of water bodies** (lakes, rivers, groundwater, oceans) by harmful substances to the extent that the water becomes unfit for human use or harmful to aquatic life.

1 mark

Three human activities causing water pollution:

- Discharge of household waste (human waste, soaps, detergents, garbage) into water bodies. ½ mark
- Agricultural runoff of pesticides, fertilizers, and livestock waste. ½ mark
- Industrial discharge (petroleum, natural gas production waste, oil leaks) into rivers and groundwater. ½ mark

(Any three clearly stated: 2 marks total for activities.)

OR

Household waste contamination: Household waste includes human wastes, livestock wastes, soaps, detergents, paints, oil, food waste, and garbage. 1 mark

When this waste is discharged into lakes, ponds, or rivers, the **chemical and bacterial content** contaminates surface and groundwater. 1 mark

Three infectious diseases: Cholera, jaundice, hepatitis, typhoid, dysentery (any three accepted). 1 mark

2(viii) Dysentery / Typhoid 3 marks

Dysentery: An intestinal disease caused by the parasite **entamoeba**. 1 mark

Spread: Transmitted by **faecal contamination** of water or food by the encysted organism. 1 mark

Symptoms: Mild to severe abdominal cramps, diarrhoea, chocolate-coloured stool with mucus and sometimes blood. 1 mark

OR

Typhoid: A dangerous intestinal disease. 1 mark

Causative bacteria: *Salmonella typhi*, *salmonella paratyphi*, and *salmonella enteritidis*. 1 mark

Characteristic fever: Continuous fever between **101°F to 104°F** and irregular pulse. 1 mark

2(ix) Ways to deal with water pollution / Heavy metals 3 marks

Any three from Section 12.5, one mark each:

- Control **nutrient flow** (nitrates and phosphates) and properly treat sewage and industrial waste to prevent oxygen depletion in water bodies. 1 mark
- Prevent **industrial wastewater** from entering waterways to minimise heavy metal emissions. 1 mark
- Ensure proper treatment of domestic and industrial wastewater; maintain good sanitation and hygiene; provide access to clean drinking water to prevent waterborne diseases. 1 mark

OR

Heavy metals such as **arsenic (As)**, **chromium (Cr)**, and **lead (Pb)** are dangerous because they **accumulate in the tissues** of aquatic organisms and pass to humans through the food chain. 1 mark

Health problems caused:

- Long-term neurological problems and nausea. 1 mark
- Kidney diseases, dizziness, and cancer (any two accepted). 1 mark

2(x) Water scarcity in Pakistan: causes and solutions 3 marks

Three causes (Section 12.6):

- **Population growth:** rapidly growing population increases demand for water in agriculture, industry, and homes. 1 mark
- **Climate change:** irregular rainfall patterns; excessive extraction depletes groundwater aquifers. 1 mark
- **Inadequate water supply:** traditional flood irrigation causes large losses through evaporation and poor distribution. 1 mark

OR

Three measures to resolve water scarcity:

- **Effective water management practices** to reduce waste. 1 mark
- **Development of infrastructure** (canals, reservoirs, modern irrigation). 1 mark
- **Policy reforms** and **public awareness** of responsible water use; strict regulations against illegal drilling. 1 mark

2(xi) Fertilizers and NPK / Nutrient management 3 marks

Definition: Fertilizers are substances that **provide essential elements for plant growth** and enhance crop yields. 1 mark

NPK nutrients and examples:

- **Nitrogen (N):** supplied by urea, ammonium salts, potassium nitrate. ½ mark
- **Phosphorus (P):** supplied by di-ammonium phosphate (superphosphate). ½ mark
- **Potassium (K):** supplied by potassium nitrate. ½ mark

OR

Nutrient management: the practice of giving fertilizers at the **right time and in the right amount** to optimise plant nutrient uptake and reduce environmental loss. 1 mark

Environmental problem: Over-use of fertilizers causes **nutrient runoff** into water bodies, where nitrates and phosphates accumulate. 1 mark

This triggers **eutrophication:** excessive algal growth, oxygen depletion, and death of aquatic life. 1 mark

3(i) Detection of water / Distilled vs. tap water comparison 5 marks

(a) Anhydrous copper(II) sulphate test: [3 marks]

Anhydrous copper(II) sulphate is a **white crystalline solid**. It does not contain water molecules but can absorb water from the surroundings through a chemical process. 1 mark

When a few drops of a water-containing sample are added, a chemical reaction occurs: it forms **hydrated copper(II) sulphate**, which is **blue**. The colour change from white to blue confirms water is present.

1 mark

Equation: $\text{CuSO}_4 + 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (white $\rightarrow$ blue) 1 mark

(b) Boiling point purity test: [2 marks]

Water is heated and the temperature at which it boils is recorded. Pure water boils at exactly **100°C** at sea level. 1 mark

A boiling point **above 100°C** indicates the presence of dissolved solids (solutes raise the boiling point). This is because dissolved impurities disrupt the escape of water molecules into vapour, requiring more energy (higher temperature). 1 mark

OR

(a) Distilled water vs. tap water [3 marks]:

Aspect	Distilled Water	Tap Water
How obtained	Boiling water to vapour, condensing back to liquid; impurities left behind	Treated municipal water from streams, rivers, lakes, reservoirs, groundwater
Dissolved solids	None; free of all contaminants	Contains dissolved minerals, trace treatment chemicals
Electrical conductivity	Very low (no ions)	Higher (dissolved minerals provide ions)
pH	7 (neutral)	Slightly variable depending on source
Lab use 1	Preparing chemical reagents and solutions	Cleaning glassware and equipment
Lab use 2	Calibrating sensors (e.g. pH meter)	Simple educational demonstrations

1 mark per three comparison points correctly stated

(b) Why natural spring water is better for drinking than distilled: [2 marks]

Long-term drinking of distilled water can cause **mineral deficiency** because it lacks essential minerals.

1 mark

Natural spring or treated tap water retains **calcium, magnesium, iron, and trace elements** needed for haemoglobin production, enzyme activation, and bone health. These cannot be obtained from distilled water. 1 mark

3(ii) Raw water treatment process / Chemical equations and public health 5 marks

(a) Four steps of raw water treatment [4 marks]:

1. **Sedimentation:** Raw water is allowed to stand in a large reservoir. Suspended solid matter (mud, silt) sinks to the bottom due to gravity. No chemicals are added at this stage. 1 mark
2. **Coagulation:** Water is treated with slaked lime $\text{Ca}(\text{OH})_2$ and alum $\text{Al}_2(\text{SO}_4)_3$. They react to form a gelatinous mass of aluminium hydroxide $\text{Al}(\text{OH})_3$, which carries down fine dirt particles and bacteria as it settles. 1 mark
3. **Filtration:** Water is filtered through beds of sand and gravel. Sometimes it is also filtered through activated charcoal to remove coloured and odorous compounds. 1 mark
4. **Chlorination:** Chlorine gas is added in a controlled amount. It reacts with water to form hypochlorous acid HClO , which kills any remaining bacteria. 1 mark

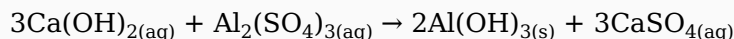
(b) Flowchart [1 mark]:

Raw Water → Sedimentation → Coagulation → Filtration → Chlorination → Pure Water

OR

(a) Chemical equations [3 marks]:

Coagulation:



The product **$\text{Al}(\text{OH})_3$** (aluminium hydroxide) is the gelatinous mass that traps dirt and bacteria. 1½ marks

Chlorination:



The product **HClO** (hypochlorous acid) is the bactericidal agent. 1½ marks

(b) Water treatment and public health [2 marks]:

Water treatment is essential because untreated water contains bacteria, viruses, parasites, and heavy metals. 1 mark

Three diseases prevented: **diarrhoea, dysentery, typhoid** (or cholera, hepatitis, polio — any three from the textbook list accepted). 1 mark

3(iii) Waterborne diseases / Water pollution sources and remedies **5 marks****(a) Four waterborne diseases [4 marks]:**

Disease	Causative organism	How it spreads	One symptom
Dysentery	Parasite <i>entamoeba</i>	Faecal contamination of water or food	Abdominal cramps, bloody diarrhoea
Jaundice	Liver obstruction (not microbe-specific in textbook)	Contaminated water	Yellowness of skin and eyes; fatigue
Hepatitis	Viruses (types A, B, C, D, E)	Drinking polluted water (types A and E)	Acute liver inflammation
Typhoid	<i>Salmonella typhi</i>	Polluted water containing the bacteria	Continuous fever 101°F to 104°F, irregular pulse

1 mark per disease correctly described (organism + spread + symptom)**(b) Prevention measures [1 mark]:**

Dysentery: ensure clean drinking water and proper food hygiene; avoid faecal contamination of water supply.

Typhoid: drink treated/boiled water; proper sanitation to prevent bacterial contamination of water sources.

1 mark**OR****(a) Three sources of water pollution and harms [3 marks]:**

- **Harmful microbes in sewage:** wastewater carries bacteria, viruses, and parasites. These cause waterborne diseases such as typhoid, cholera, and hepatitis in humans. **1 mark**
- **Metal compounds:** toxic metals (As, Cr, Pb) accumulate in aquatic organism tissues and pass to humans, causing neurological damage, kidney disease, and cancer. **1 mark**
- **Nitrates and phosphates from fertilizers:** cause eutrophication; excessive algal blooms deplete oxygen in water bodies, killing fish and other aquatic life. **1 mark**

(b) Three ways to deal with negative effects [2 marks]:

- Control nutrient flow and properly treat sewage and industrial waste to prevent oxygen depletion. **½ mark**
- Prevent industrial wastewater from entering waterways to minimise heavy metal pollution. **½ mark**
- Ensure proper treatment of domestic and industrial wastewater; maintain sanitation and hygiene; provide access to clean drinking water. **1 mark**

3(iv) Water scarcity in Pakistan / NPK fertilizers and eutrophication 5 marks

(a) Causes and solutions of water scarcity in Pakistan [4 marks]:

Water scarcity is a **major problem in Pakistan** because demand exceeds the available safe water supply.

1 mark

Three causes:

- **Population growth:** Pakistan's rapidly growing population increases water demand in agriculture, industry, and households. 1 mark
- **Climate change:** irregular rainfall patterns and excessive groundwater extraction deplete aquifers; country's water resources are declining. 1 mark
- **Inadequate water supply:** traditional flood irrigation causes significant losses through evaporation and poor water distribution. 1 mark

Three solutions: (i) effective water management practices; (ii) development of infrastructure; (iii) policy reforms + public awareness of responsible water use + strict regulation of groundwater drilling.

(b) Climate change and water scarcity [1 mark]:

Climate change causes **irregular rainfall patterns**, reducing water availability. The country's water resources decline because of **excessive extraction from groundwater aquifers** and **insufficient irrigation practices**. This means less water recharges underground reserves. 1 mark

OR

(a) NPK fertilizers [3 marks]:

Definition: Fertilizers are substances that provide essential elements for plant growth to enhance crop yields. 1 mark

NPK stands for **nitrogen (N), phosphorus (P), and potassium (K)**. These three main nutrients are provided by:

- **Nitrogen:** urea, ammonium salts (e.g. ammonium nitrate), potassium nitrate. ½ mark
- **Phosphorus:** di-ammonium phosphate, superphosphate. ½ mark
- **Potassium:** potassium nitrate. ½ mark

Fertilizers dissolve in water, making nutrients available in a readily usable form for plants. ½ mark

(b) Excessive fertilizers and eutrophication [2 marks]:

When fertilizers are used in excess, **nitrates and phosphates run off** into nearby rivers, lakes, and water bodies through agricultural drainage. 1 mark

This causes **eutrophication**: phosphates trigger explosive growth of algae (algal blooms). The algae consume all available oxygen in the water. The resulting **oxygen depletion kills fish and other aquatic organisms**. 1 mark